Who Matters Most in a Citation Network?

- Citation network analysis of the bibliometrics literature
- Introduction to Network Science
- [YOUR NAME] | ID: [STUDENT ID]
- October 28, 2026

Research questions

- Main question: which papers are most influential in the bibliometrics literature, and how are they connected?
- Q1. Which papers receive the most incoming citations (in-degree)?
- Q2. Which papers act as bridges between parts of the network (betweenness)?
- Q3. Does PageRank re-order the in-degree ranking?
- Q4. What communities exist, and do rankings change over time?

Data and methods

- Source: OpenAlex API, retrieved 2026-10-20; no login needed.
- Sample: top-cited journal articles whose title matches “bibliometric”, “citation analysis”, “co-citation”, “bibliographic coupling”, “scientometrics” (up to 100 per query), merged, de-duplicated by title, top 150 kept.
- Edge A → B means A cites B. Only citations between sampled papers are kept.
- Cleaning: 4 self-loops removed; 1 edge points to a later-published paper (metadata artefact, kept and flagged).
- Result: N = 150 papers, E = 424 citations; 133 papers lie in the largest weak component.
- Tools: Python (NetworkX, pandas, SciPy, Matplotlib). Communities: Louvain on the undirected graph (seed 42).
- Clustering is computed on the undirected graph; path length only on the largest weak component (also undirected).

Network metrics

Metric	Value
Papers (N)	150
Citations (E)	424
Mean in-degree (E/N)	2.83
Mean total degree (2E/N)	5.65
Density E/[N(N−1)]	0.0190
Papers with no links	15
Weak components	17
Average clustering (undirected)	0.212
Avg. path length, largest component	2.92
Diameter, largest component	7
Louvain modularity (24 groups)	0.421

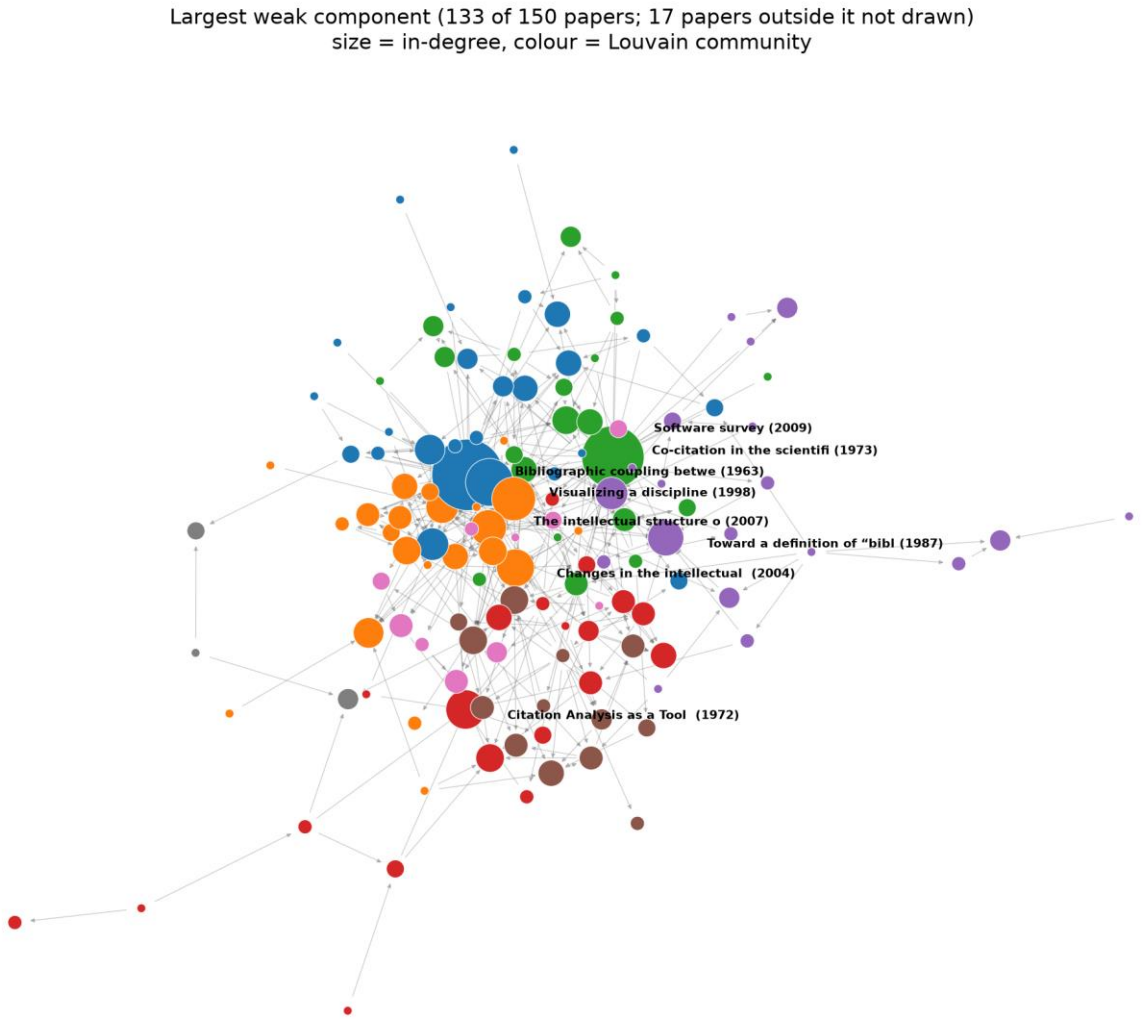
- 33% of papers are never cited inside the sample.
- Maximum in-degree is 41 against a mean of 2.83: citations are highly concentrated.
- The network is sparse: density is under 2%.

Most-cited papers inside the network (in-degree)

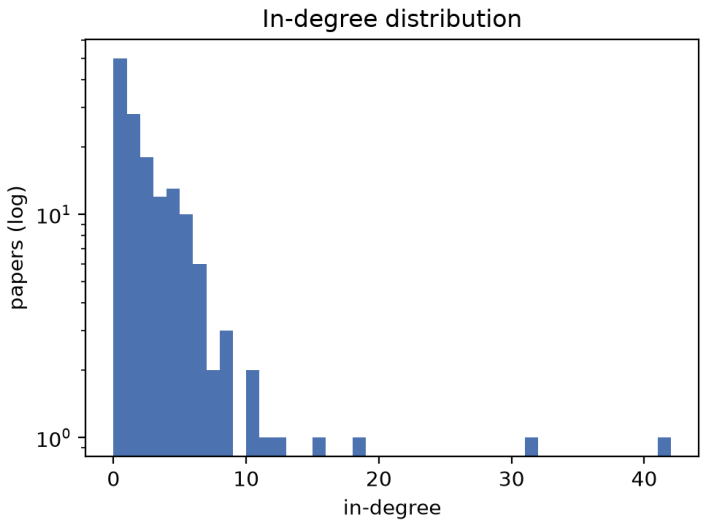
Rank	Paper	Year	In-degree
1	Co-citation in the scientific literature: A new measure of the relationship b…	1973	41
2	Software survey: VOSviewer, a computer program for bibliometric mapping	2009	31
3	Bibliographic coupling between scientific papers	1963	18
4	Visualizing a discipline: An author co-citation analysis of information scien…	1998	15
5	Citation Analysis as a Tool in Journal Evaluation	1972	12
6	Changes in the intellectual structure of strategic management research: a bib…	2004	11
7	Toward a definition of “bibliometrics”	1987	10
8	The intellectual structure of the strategic management field: an author co-ci…	2007	10
9	The Literature of Bibliometrics, Scientometrics, and Informetrics	2001	8
10	Bibliometric Methods in Management and Organization	2014	8

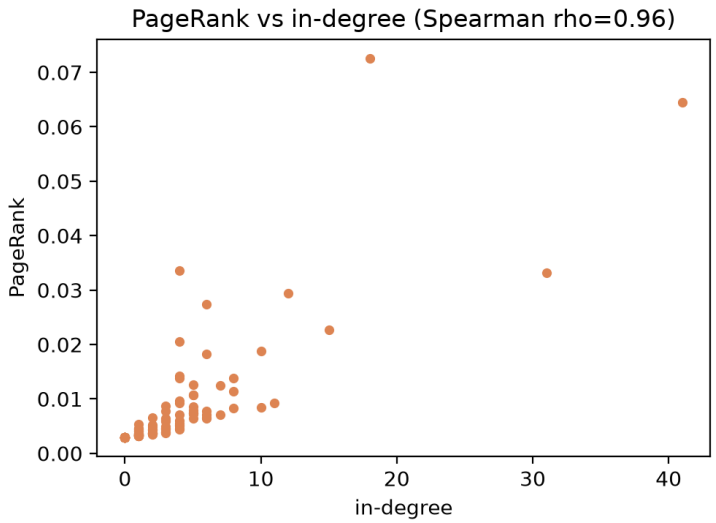
In-degree counts citations from the other 149 sampled papers only. Rank correlation with global OpenAlex citation counts: Spearman $\rho = 0.35$.

The citation network



In-degree versus PageRank





PR	Top PageRank papers	Year
1	Bibliographic coupling between scientif…	1963
2	Co-citation in the scientific literatur…	1973
3	From the science of science to Scientom…	2009
4	Software survey: VOSviewer, a computer …	2009
5	Citation Analysis as a Tool in Journal …	1972
6	Scholarly communication and bibliometri…	2002
7	Visualizing a discipline: An author co-…	1998
8	The use of bibliometric data for the me…	1985
9	Toward a definition of “bibliometrics”	1987
10	New bibliometric tools for the assessme…	1995

- Spearman ρ (in-degree, PageRank) = 0.96: the two rankings broadly agree.
- But only 6 of the top 10 are shared; PageRank rewards being cited by well-cited papers.
- Older papers collect more in-sample citations (Spearman ρ year vs in-degree = -0.61).

Bridge papers (betweenness centrality)

Rank	Paper	Year	Betweenness
1	How to conduct a bibliometric analysis: An overview and guidelines	2021	0.0109
2	Bibliometric Methods in Management and Organization	2014	0.0071
3	Scholarly communication and bibliometrics	2002	0.0071
4	The intellectual structure of the strategic management field: an author co-cita…	2007	0.0052
5	Changes in the intellectual structure of strategic management research: a bibli…	2004	0.0045
6	Visualizing a discipline: An author co-citation analysis of information science…	1998	0.0044
7	Bibliometric Methods: Pitfalls and Possibilities	2005	0.0042
8	Forty-five years of Journal of Business Research: A bibliometric analysis	2019	0.0038
9	A bibliometric analysis of operations research and management science	2016	0.0036
10	The evolution of the intellectual structure of operations management—1980–2006:…	2008	0.0031

Directed betweenness. Top bridges are mostly review/method papers that cite across fields; foundational papers are cited a lot but sit at path endpoints, so they score low.

Communities (Louvain)

Group	Papers	Median year	Most-cited paper in group (in-sample)
0	25	2015	Co-citation in the scientific literature: A n (1973)
1	22	2007	Visualizing a discipline: An author co-citati (1998)
2	21	2016	Software survey: VOSviewer, a computer progra (2009)
3	20	2014	Citation Analysis as a Tool in Journal Evalua (1972)
4	19	2018	Toward a definition of “bibliometrics” (1987)
5	13	2006	Bibliometric Methods: Pitfalls and Possibilit (2005)
6	10	2015	The Structural Influence of Marketing Journal (2003)
7	3	2015	Bibliometrics: The Leiden Manifesto for resea (2015)

Modularity 0.421 with 8 groups of 3+ papers; remaining groups are single papers with no in-sample links. Group labels are computed only on structure, not on topic.

Rankings through time (network up to 2010 vs full)

Paper (year)	Rank ≤2010	Rank full	In-deg ≤2010	In-deg full
Co-citation in the scientific literature: A n (1973)	1	1	14	41
Citation Analysis as a Tool in Journal Evalua (1972)	2	5	7	12
Visualizing a discipline: An author co-citati (1998)	2	4	7	15
Bibliographic coupling between scientific pap (1963)	4	3	5	18
Scholarly communication and bibliometrics (2002)	4	14	5	6
Mapping of science by combined co-citation an (1991)	6	9	4	8
Is citation analysis a legitimate evaluation (1979)	6	12	4	7
Problems of citation analysis (1996)	6	30	4	4
New bibliometric tools for the assessment of (1995)	6	14	4	6
The Intellectual Development of Management In (1986)	10	20	3	5

Early network: 61 papers, 93 citations. Ties share a rank. The 1973 co-citation paper stays first; several early papers fall in the full ranking as later papers cite other work.

Limitations

- Database: OpenAlex coverage and reference lists differ from Web of Science or Scopus.
- Sampling: title keyword search ranked by global citations; applied bibliometric studies (e.g. in management) dominate the recent half and some topics are missed.
- Closed sample: only citations among the 150 papers count, so in-degree is not total citations.
- Citation age: older papers had longer to be cited; self-citations are not removed.
- Single run: Louvain is stochastic (seed fixed) and the cutoff year is a choice.
- Citation counts measure attention, not quality.

Conclusions

- Most-cited inside the network: “Co-citation in the scientific literature: A new measure of the relati…” (1973), in-degree 41.
- Influence depends on the measure: in-degree, PageRank and betweenness pick out different papers (foundational, well-connected, and bridging).
- Structure: sparse (density 0.019) but 133 of 150 papers are connected; modularity 0.42 indicates moderate community structure.
- Citation counts alone do not measure quality, but network position shows the structural role each paper plays.

References and data

- OpenAlex API (https://openalex.org); Priem, Piwowar & Orr (2022), OpenAlex: a fully-open index of scholarly works.
- Hagberg, Schult & Swart (2008). Exploring network structure, dynamics, and function using NetworkX.
- Blondel et al. (2008). Fast unfolding of communities in large networks. J. Stat. Mech.
- Page, Brin, Motwani & Winograd (1999). The PageRank citation ranking.

- Reproducibility: data/processed/{nodes,edges}.csv and metadata.json; src/analyze.py regenerates every figure and number.